# Physics
# Higher level
# Paper 2
# B.3 Gas laws

12 October 2026

**W4ss Academy** Paper 2 practice 2 | **Topic** B.3 Gas laws | **Ref.** W4SS–PHY–B3–P2–02

Student name

35 minutes

**Instructions to students**

- Write your name in the boxes above.
- Do not open this examination paper until instructed to do so.
- Answer all questions.
- Answers must be written within the answer boxes provided.
- A calculator is required for this paper.
- A clean copy of the **physics data booklet** is required for this paper.
- The maximum mark for this examination paper is **[20 marks]**.

Answer **all** questions. Answers must be written within the answer boxes provided.

**1.** This question is about a nuclear power station.

In the reactor, uranium-235 nuclei undergo fission. One possible fission reaction is

$$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{132}_{52}Te + {}^{100}_{40}Zr + 4\ {}^{1}_{0}n$$

(a) (i) State the principal energy change in nuclear fission. [1]

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(ii) The energy released in the reaction is about 180 MeV. Estimate, in J, the energy released when 1 kg of $^{235}_{92}U$ undergoes fission. [3]

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Tellurium-132 is one of the products of this reaction. It is radioactive.

(b) (i) State and explain the decay mode of $^{132}_{52}Te$ . [2]

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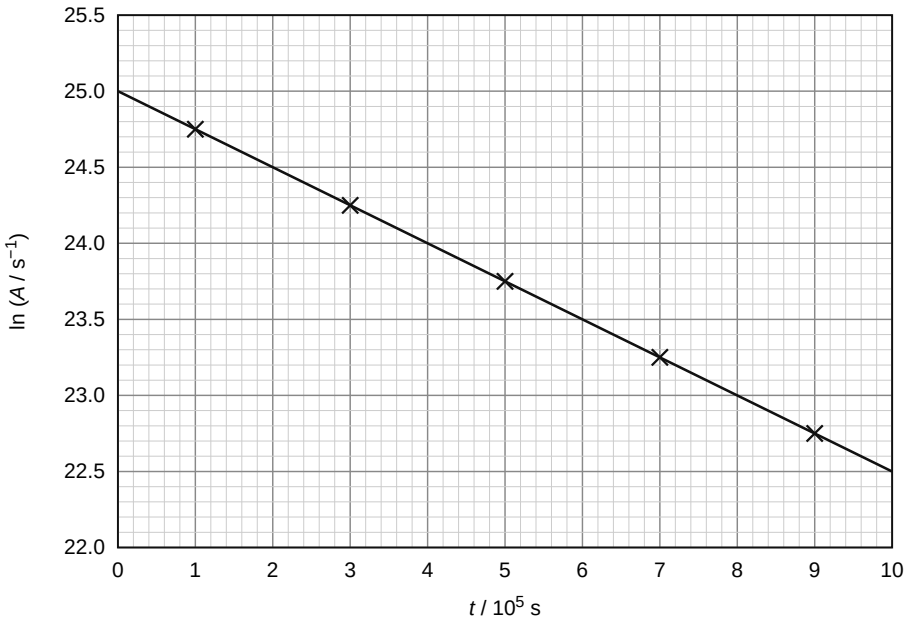
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**(Question 1 continued)**

A sample of tellurium-132 is removed from the reactor. The graph shows the variation with time $t$ of ln $A$, where $A$ is the activity of the sample in $s^{-1}$.




(ii) Calculate, in $s^{-1}$, the initial activity of the sample. [1]

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(iii) Show that the decay constant of a nuclide is given by $\lambda = -m$, where $m$ is the slope of the graph of ln $A$ against $t$. [1]

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**(Question 1 continued)**

(iv) Determine, in days, the half-life of $^{132}_{52}Te$ . [2]

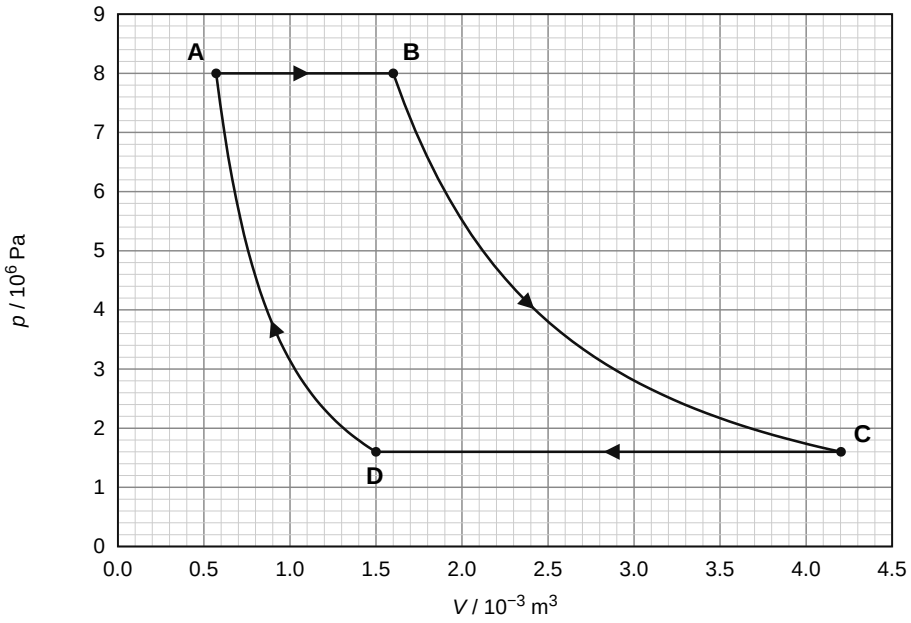
In this power station, the reactor is cooled by a liquid coolant. A heat exchanger transfers thermal energy from the coolant to helium gas, which drives a turbine.

(c) (i) Outline the role of the heat exchanger in a nuclear power station. [1]

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**(Question 1 continued)**

1.0 mol of helium, which can be treated as an ideal monatomic gas, is taken through the cycle ABCDA shown. AB and CD are isobaric changes. BC and DA are adiabatic changes.




(ii) Calculate the maximum temperature of the gas during the cycle. [3]

(iii) Outline why the entropy of the gas remains constant during changes BC and DA. [1]

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**(Question 1 continued)**

The table gives the work done by the gas during each change of the cycle. During AB, 20.58 kJ of thermal energy is transferred to the gas.

| Change | Work done by the gas / kJ |
|---|---|
| A → B | 8.23 |
| B → C | 9.11 |
| C → D | −4.32 |
| D → A | −3.25 |

(iv) Determine the efficiency of the cycle. [2]

When the power station is shut down, the turbine slows down. The table shows the angular speed $\omega$ of the turbine at different times $t$ after the shutdown.

| $t$ / s | 0 | 20 | 40 | 60 | 80 | 100 |
|---|---|---|---|---|---|---|
| $\omega$ / rad s$^{-1}$ | 310 | 277 | 240 | 196 | 139 | 0 |

(d) Show that the rotational kinetic energy of the turbine decreases at a constant rate. [3]

## Sources of the questions

This practice test was put together by W4ss Academy for internal class use. The questions are taken from the IB DP Physics Questionbank (first assessment 2025), topic B.3 Gas laws. They remain the property of the International Baccalaureate Organization and must not be published or sold.

The introductions of these questions are missing from the Questionbank. W4ss Academy wrote them using only the data of the official markscheme, so that the markscheme still applies.

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|---|---|
| Paper 2 — question 1 | EXE.2.HL.TZ0.1 — introduction written by W4ss Academy from the markscheme (graphs and tables drawn by W4ss Academy; “λ =” restored in (b) (iii)) |

Please **do not** write on this page.

Answers written on this page
will not be marked.